Raffaele Castania

raffaele.castania.04@gmail.com | (+39) 3517223724 | Via dei giardini 4, Tremestieri Etneo (CT)
https://github.com/0pow-pow0 | Date of Birth: 10 June 2004 | Portfolio (Click me!)

Skills

English (B2), C#, Unity Profiler, Frame Debugger, Animator, Blend Trees, Animation Layers, IK, Physics Queries, Raycast, Rigidbody, Coroutines, ScriptableObject, Cinemachine, DOTween, LOD, Linear Algebra, Quaternions, Vectors, Matrices, Input System, UI Toolkit, UXML, USS, URP, Post-Processing, Git, GitHub, Gitflow, SOLID Principles, Design Patterns, Timeline, UML.

Profile

Passionate about video games since childhood, I am a curious and proactive individual with a strong focus on problem-solving and continuous growth. Enthusiastic and team-oriented, I am comfortable sharing ideas and taking responsibility when necessary.

Project Experience (Rationale behind the solutions available in the Portfolio)

Rumble Smash - Unity C# 12/02/2026 - 26/05/2026

- Developed a Dialogue System and a Quest System using ScriptableObjects, separating data from logic to enable efficient management of 30+ assets. Integrated the systems with Unity Timeline, allowing cutscenes, dialogues, and quests to be layered while maintaining full control over the execution flow through the Inspector and eliminating the need for boilerplate code.
- Developed a Finite State Machine capable of managing 4+ states of an entity, reducing conditional complexity and enabling efficient management of 40+ variables related to physics, rendering, and gameplay logic. Implemented an event-driven system to provide complete control over Rigidbody variables based on the entity's current state.
- Designed a single-entry-point system for managing the match flow, coordinating transitions between the pre-match interface, device assignment, match initialization, and the retry interface at the end of the match. Implemented an event-driven architecture to centralize event sequence management, simplifying the integration and handling of the different stages of the game cycle.
- Developed animation systems using Animator Controllers, Avatars, Avatar Masks, Blend Trees, Animation Clips, and parameters, managing transitions between animation states and integrating gameplay logic to synchronize visual feedback with the game context.
- Utilized DOTween, Tweeners, Sequences, and Callbacks in combination with Coroutines to implement animations and visual feedback for UI and gameplay elements, making interactions clearer, more responsive, and more intuitive for the player.

Other Projects - Unity C# 04/05/2025 - 2/02/2026

- Developed an Occlusion Detection algorithm for the player's camera, detecting obstacles using Raycasts and dynamically modifying the transparency of the corresponding materials to keep the player visible. - *Identity Seeker*
- Developed a Projectile-Based Shooting system, in which weapons generate independent projectiles with their own lifecycle, behavior, and collision handling. Configured different projectile types through data-driven parameters, enabling the definition of specific values for speed, damage, and effects. Managed collision handling through LayerMasks, while keeping the shooting, projectile, and damage systems decoupled to reduce tight coupling and direct references. - *Chicken Revenge*
- Developed a system for dynamically assigning input devices to 2+ players using the Unity Input System and the PlayerInput component. Implemented a dedicated interface allowing players to intuitively configure and verify their assigned devices. - *SSGRP*
- Developed an algorithm for collision detection for entities within a grid using C# Lists, separating the translation axes of an entity (the tetramino) and analyzing them independently, predicting its translation based on player input and resolving collisions according to the context of the grid. - *PuyoTetris (Clone)*
- Implemented a data-driven power-up system supporting 10+ statistics and 6+ special effects, leveraging ScriptableObjects and applying the Strategy Pattern to encapsulate different behaviors for each power-up. - *Chicken Revenge*
- Developed an AI system based on a Finite State Machine, integrating it with the NavMeshAgent component for movement and implementing dedicated states for different behaviors and their corresponding conditional transitions. This enabled dynamic management of the entity's behavior during gameplay while keeping the decision-making logic modular and extensible. - *Toyorun*

Education

Game Developer Generalist Certificate 19/09/2024 - 16/06/2026
Nautilus Academy · Via Monserrato, 110, 95128 Catania CT

I authorize the processing of the personal data contained in my curriculum vitae in accordance with EU Regulation 2016/679 (GDPR) and applicable Italian legislation.